

BEN MEDCALF

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PROFESSIONAL SUMMARY

Quality engineer with over a decade of experience in root cause analysis, data validation, and systematic debugging across enterprise systems. Recently expanded into graph database consulting, applying relationship-modeling techniques to complex data problems. Background combines deep technical investigation skills with strong documentation practices and cross-functional collaboration.

TECHNICAL SKILLS

Quality & Analysis: Root Cause Analysis, Data Validation, Test Strategy Design, Defect Investigation, Technical Documentation

Databases: SQL, Neo4j (Graph Databases), Cypher Query Language, Data Modeling, ETL Pipelines

Programming: Python, JavaScript/TypeScript, Kotlin, Java, C#, REST APIs

Cloud & DevOps: Azure (CosmosDB, AppServices, KeyVault, AppInsights), Docker, CI/CD Automation

PROFESSIONAL EXPERIENCE

Senior Software Development Engineer in Test (SDET)

Enterprise Software Companies • 2015 – 2025

- Performed systematic root cause analysis on complex defects, tracing issues across microservice architectures and database layers to identify failure points
- Developed and executed data validation strategies for enterprise applications handling sensitive customer financial data
- Built automated testing frameworks and CI/CD pipelines to catch data integrity issues before production deployment
- Documented defects and root causes for engineering teams, enabling development of permanent fixes
- Collaborated with developers, DBAs, and product teams to prioritize and resolve issues in production systems

Founder & Principal Consultant

Medcalf Software Solutions, LLC • May 2025 – Present

- Designed data extraction and validation pipeline for legal documents, implementing confidence scoring to flag ambiguous results for human review
- Built graph database solutions using Neo4j to model relationships between entities across multiple data sources
- Created technical documentation enabling engineering teams to build production systems from proof-of-concept work

Founder & Lead Engineer

StatFoundry.com • May 2025 – Present

- Architected graph database to model relationships across 5+ years of data from multiple sources
- Developed ETL pipeline for ingesting and validating data from disparate tabular sources into unified structure

EDUCATION & CERTIFICATIONS

Neo4j Certified Professional | **Knowledge Graph Builders Certificate** | **React Nanodegree** (Udacity)